

Address APIs User - Guide

FOR NATIONAL ADDRESS DATA API'S PROJECT

National Address integration APIs, June 08 2023



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Introduction

This document explains the National Address (NA) APIs and the way it can be consumed by National Address clients. The NA APIs creation mainly derived from our agreement and understanding of the business process and the required data needed to be exchange in order to satisfy the business need for all parties involved in this project.

In this document:

MUST means that the requirement is an absolute mandatory requirement.

MUST NOT means that the requirement absolutely prohibited.

SHOULD means that if the requirement cannot be fulfilled out-of-the-box, some other means of fulfilling the requirement should be devised, e.g. by customization or by integrating with another product that can fulfill the functionality.

MAY means that the requirement is optional.

National Address Integration Services (APIs)

Address APIs

Service Description

Full National Address By Geocode:

Full National Address By Geocode allows a user to find the detailed National Address including Short Address for a given Geocode. You can decide to use Json or XML as an output format in addition to language of your choice from Arabic or English.

Service URLs

Production Environment:

https://apina.address.gov.sa/NationalAddress/FullNationalAddressByGeocode/FullNationalAddressByGeocode?format=xml&lat=24.787489&long=46.619769&api_key=<apikey>

Service Methods

Full National Address By Geocode *Method(GET)*

Input Parameters

Parameter Name	Mandatory	Data Type	Description
lat	Mandatory	String	Latitude in decimal units. Latitude must be between -90 and +90
long	Mandatory	String	Longitude in decimal units. Longitude must be between -180 and +180
language	Optional ('A' or 'E')	String	It is an optional input. Our API can provide response in Arabic or English with Arabic being the default language.
format	Optional ('json' or 'xml')	String	It's an optional input. We currently support output for API's in two formats Json and XML. You will be required to specify in which format you need response, default is Json.

encode	Optional (‘utf8’ or ‘windows1256’)	String	By default response is in windows-1256. Pass "utf8" as input parameter to get utf-8 charset.
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Sample Request:

<https://apina.address.gov.sa/NationalAddress/FullNationalAddressByGeocode/FullNationalAddressByGeocode?format=json&language=E&encode=utf8&lat=24.68000404&long=46.70236983>

Output Parameters

Parameter Name	Mandatory	Data Type	Description
NationalAddress	Mandatory	Object	
Restriction	Mandatory	String	Restriction: indicates restriction status (1 = true, 2 = false). In case value is empty, default value will be '2' (it means that no restrictions apply to the address.)
ShortAddress	Mandatory	String	The short address.
Address1	Mandatory	String	This consists of the first line of National Address format of address." [BuildingNumber] [Street] - [District]".
Address2	Mandatory	String	This consists of the second line of National Address format of address." [City] - [PostCode] – [AdditionalNumber]".
ObjLatLng	Mandatory	String	This consists of latitude and longitude details of the address. To enable user to plot it on any map application like google map. [ObjectId] [Longitude] [Latitude] [Minx] [Miny] [Maxx] [Maxy].

RegionName	Mandatory	String	Name of region in which address falls. Example Ar Riyadh.
PKAddressID	Mandatory	String	Primary key for address. Example : 1264382282121
DistrictID	Optional	String	District Id in which address falls. Example : 10100003001
CityId	Mandatory	String	Id of the city in which address falls. Example: 3
RegionId	Mandatory	String	Id of the region in which address falls. Example 1
Street_L2	Mandatory	String	Street name in which address falls with another available language from default.
District_L2	Mandatory	String	Street name in which address falls with another available language from default.
City_L2	Mandatory	String	City name in which address falls with another available language from default.
RegionName_L2	Mandatory	String	Region name in which address falls with another available language from default.
BuildingNumber	Mandatory	String	Four digit number which is unique to a building.
PostCode	Mandatory	String	Four digits numerical Post code / Zip Code.
AdditionalNumber	Mandatory	String	Four digit number.
Street	Mandatory	String	Name of the street on which building is. Example king abdul aziz road.
District	Mandatory	String	Name of the district in which address falls. Example al amal.

City	Mandatory	String	Name of the city in which address falls. Example Riyadh.
success	Mandatory	Boolean	This represents success or failure of the request itself (true/false)
Message	Optional	Object	
MessageCode	Optional	String	Unique identifier or code representing the message.
MessageDescriptionEn	Optional	String	Description of the error message in Arabic
MessageDescriptionAr	Optional	String	Description of the error message in Arabic

Sample Response 1 (Success case):

```
{
  "NationalAddress": {
    "Restriction": "2",
    "ShortAddress": "MBDA2115",
    "Address1": "2115 Saha Islam 1 St - Al Bibyan Dist.",
    "Address2": "MAKKAH 24231 - 8921",
    "ObjLatLng": "39.80535713 21.43763644",
    "RegionName": "Makkah",
    "PKAddressID": "2423121158921",
    "DistrictID": "10200006027",
    "CityId": "6",
    "RegionId": "2",
    "Street_L2": "شارع 1 ساحة اسلام",
    "District_L2": "حي البيبان",
    "City_L2": "مكة المكرمة",
    "RegionName_L2": "منطقة مكة المكرمة",
    "BuildingNumber": "2115",
    "PostCode": "24231",
    "AdditionalNumber": "8921",
    "Street": "Saha Islam 1 St",
    "District": "Al Bibyan Dist.",
    "City": "MAKKAH"
  },
  "success": true,
```

```
"Message": null
}
```

Sample Response 2 (Failure case):

```
{
  "NationalAddress": null,
  "success": false,
  "Message": {
    "MessageCode": "MSG001",
    "MessageDescriptionEn": "Something went wrong, please try again later",
    "MessageDescriptionAr": "حدث خطأ ما، الرجاء إعادة المحاولة لاحقاً"
  }
}
```

Sample Response 3 (Failure case):

```
{
  "NationalAddress": null,
  "success": false,
  "Message": {
    "MessageCode": "MSG002",
    "MessageDescriptionEn": "Invalid format, please try again",
    "MessageDescriptionAr": "الادخال غير صحيح، الرجاء إعادة المحاولة"
  }
}
```

Sample Response 4 (Failure case):

```
{
  "NationalAddress": null,
  "success": true,
  "Message": {
    "MessageCode": "MSG005",
    "MessageDescriptionEn": "The selected location does not refer to a building, please try again",
    "MessageDescriptionAr": "الموقع المحدد لا يشير الى مبني، الرجاء إعادة المحاولة"
  }
}
```

Error Handling

HTTP Status Code	Scenario	Error Code
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200	The latitude or longitude provided is not a valid number or out of range.	MSG001
200	The latitude or longitude field is empty when it is required.	MSG002
200	The selected location is not a building.	MSG005
401	Access denied due to invalid subscription key.	
403	Total limit for calls under your subscription has been reached.	
404	The endpoint or resource your application is trying to reach doesn't exist.	
500	There was an error on the server while processing the request.	
503	Service unavailable.	